Siyu Yuan

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RESEARCH INTEREST

Siyu Yuan（员司雨）is a third-year Ph.D. student at Fudan University in the School of Data Science. Her interested research topics are mostly around cognitive science with generative agents, including (but are not limited to):

- **Cognitive reasoning**, especially on exploring the cognitive reasoning abilities of generative agents, including *analogical reasoning*, *concept understanding*, *role-playing* and *belief exploration* of language agents. The ultimate goal is to enhance the understanding of these agents about themselves and others, thereby enabling them to generate responses that align better with human cognition.
- **Strategic Planning**, especially on equipping generative agents with human-level planning capabilities revolving around *constrained planning*, *tool invocation* and *multitasking planning*.
- **Knowledge Acquisition**, especially on excavating knowledge based on generative agents, including *concept acquisition*, *script generation*, *idiom construction*, *pun generation* and *analogy making*. These aim to construct rich knowledge resources that can be effectively utilized.

EDUCATION

Ph.D. in Statistics. **Fudan University**
Knowledge Works Laboratory at Fudan University, Advisor: Prof. Deqing Yang. *2021-2026 (estimated)*

B.S. in Data Science. **Fudan University**
Knowledge Works Laboratory at Fudan University, Advisor: Prof. Deqing Yang. *2017-2021*

EXPERIENCE

Microsoft Research Asia (Shanghai) *September 2023 - Present*
- Role: Research Intern
- Advisor: Dr. Kaitao Song
- Topics: Autonomous Agents with Planning and Tool Use
- Project: Participate in the JARVIS project (Github Stars: 22.1K) [2] [6] [20]

ByteDance AI-Lab (Shanghai) *January 2023 - May 2023*
- Role: Research Intern
- Advisor: Dr. Changzhi Sun and Dr. Jiaze Chen
- Topics: Evaluation and Improvement on the analogical reasoning abilities of LLMs [7].

Brain Technologies, NLP Research Intern (Remote) *June 2022 - September 2022*
- Role: Research Intern
- Advisor: Dr. Charles Jankowski
- Topics: Symbolic Knowledge Distillation for Language Planning and LLM Prompt Engineering [10].

Knowledge Works Lab at Fudan University (Shanghai) *March 2019 - Present*
- Role: Student Researcher
- Advisor: Prof. Yanghua Xiao and Dr. Jiangjie Chen
- Topics: Knowledge Acquisition, e.g, Analogy-making [9], Entity Typing [12], and Concept Extraction [13] [11].

AWARDS AND ACHIEVEMENTS

ACL 2023 Outstanding Paper Award (Top 1%) *2023*
- Association of Computational Linguistics

Huatai Securities Scholarship *2023*
- Fudan University

Outstanding Graduate Student of Shanghai Colleges and University (Top 5%) *2021*
- Fudan University

Outstanding Student pacemaker of Fudan University *2021*
- Fudan University

China National Scholarship *2018*
- Fudan University

SELECTED PUBLICATIONS

[*: Equal Contribution; ♠: Student I Mentored; †: Project Leader;]

[1] Boosting Scientific Concepts Understanding: Can Analogies from Teacher Models Empower Student Models?
Siyu Yuan*, Jiayang Cheng*, Lin Qiu, Deqing Yang
Short paper, Under Review, 2024.

[2] EvoAgent: Towards Multi-Agent Generation via Evolutionary Algorithms
Siyu Yuan*, Kaitao Song*, Jiangjie Chen, Xu Tan, Dongsheng Li, Deqing Yang
Long paper, Under Review, 2024.

[3] "A good pun is its own reword": Can Large Language Models Understand Puns?
Zhijun Xu*♠, **Siyu Yuan***†, Lingjie Chen♠, Deqing Yang
Long paper, Preprint, 2024.

[4] From Persona to Personalization: A Survey on Role-Playing Language Agents
Jiangjie Chen*, Xintao Wang*, Rui Xu*, **Siyu Yuan***, Yikai Zhang*♠, Wei Shi*, Jian Xie*, Shuang Li♠, Ruihan Yang♠, Tinghui Zhu, Aili Chen, Nianqi Li♠, Lida Chen, Caiyu Hu, Siye Wu, Scott Ren, Ziquan Fu, Yanghua Xiao
Long paper, Preprint, 2024.

[5] Put Your Money Where Your Mouth Is: Evaluating Strategic Planning and Execution of LLM Agents in an Auction Arena
Jiangjie Chen, **Siyu Yuan**, Rong Ye, Bodhisattwa Prasad Majumder, Kyle Richardson
Long paper, Preprint, 2024.

[6] EASYTOOL: Enhancing LLM-based Agents with Concise Tool Instruction
Siyu Yuan*, Kaitao Song*, Jiangjie Chen, Xu Tan, Yongliang Shen, Ren Kan, Dongsheng Li, Deqing Yang
Long paper, in: **ICLR 2024 Workshop on LLM Agents**.

[7] AnalogyKB: Unlocking Analogical Reasoning of Language Models with A Million-scale Knowledge Base
Siyu Yuan*, Jiangjie Chen*, Jiaqing Liang, Changzhi Sun, Yanghua Xiao, Deqing Yang
Long paper, in: **ACL 2024**.

[8] TIMEARENA: Shaping Efficient Multitasking Language Agents in a Time-Aware Simulation
Yikai Zhang♠, **Siyu Yuan**†, Caiyu Hu, Kyle Richardson, Yanghua Xiao, Jiangjie Chen
Long paper, in: **ACL 2024**.

[9] Beneath Surface Similarity: Large Language Models Make Reasonable Scientific Analogies after Structure Abduction
Siyu Yuan, Jiangjie Chen, Xuyang Ge♠, Yanghua Xiao, Deqing Yang
Long paper, in: **EMNLP 2023** (Findings).

[10] Distilling Script Knowledge from Large Language Models for Constrained Language Planning
Siyu Yuan, Jiangjie Chen, Ziquan Fu, Xuyang Ge♠, Soham Shah, Charles Robert Jankowski, Yanghua Xiao, Deqing Yang
Long paper, in: **ACL 2023**. **Outstanding Paper Award** (Top 1%)

[11] Causality-aware Concept Extraction based on Knowledge-guided Prompting
Siyu Yuan, Deqing Yang, Jinxi Liu, Shuyu Tian, Jiaqing Liang, Yanghua Xiao, Rui Xie
Long paper, in: **ACL 2023**.

[12] Generative Entity Typing with Curriculum Learning
Siyu Yuan, Deqing Yang, Jiaqing Liang, Zhixu Li, Jinxi Liu, Jingyue Huang, Yanghua Xiao
Long paper, in: **EMNLP 2022**.

[13] Large-Scale Multi-granular Concept Extraction Based on Machine Reading Comprehension
Siyu Yuan, Deqing Yang, Jiaqing Liang, Jilun Sun, Jingyue Huang, Kaiyan Cao, Yanghua Xiao, Rui Xie.
Long paper, in: **ISWC 2021**.

COMPLETE PUBLICATIONS

[*: Equal Contribution; ♠: Student I Mentored; †: Project Leader;]

[14] SELFGOAL: Your Language Agents Already Know How to Achieve High-level Goals
Ruihan Yang♠, Jiangjie Chen, Yikai Zhang♠, **Siyu Yuan**, Kyle Richardson, Yanghua Xiao, Deqing Yang
Long paper, Preprint, 2024.

[15] MINDECHO: Role-Playing Language Agents for Key Opinion Leaders
Rui Xu, Dakuan Lu, Xiaoyu Tan, Xintao Wang, **Siyu Yuan**, Jiangjie Chen, Wei Chu, Xu Yinghui
Long paper, Under Review, 2024.

[16] Evaluating Character Understanding of Large Language Models via Character Profiling from Fictional Works
Xinfeng Yuan♠, **Siyu Yuan**†, Yuhan Cui*♠, Tianhe Lin♠, Xintao Wang, Rui Xu, Jiangjie Chen, Deqing Yang
Long paper, Preprint, 2024.

[17] Character is Destiny: Can Large Language Models Simulate Persona-Driven Decisions in Role-Playing?
Rui Xu, Xintao Wang, Jiangjie Chen, **Siyu Yuan**, Xinfeng Yuan♠, Jiaqing Liang, Zulong Chen, xiaoqingdong, Yanghua

Xiao
Long paper, Preprint, 2024.

[18] InCharacter: Evaluating Personality Fidelity in Role-Playing Agents through Psychological Interviews
Xintao Wang, Yunze Xiao, Jen-tse Huang, **Siyu Yuan**, Rui Xu, Haoran Guo, Quan Tu, Yaying Fei, Ziang Leng, Wei Wang, Jiangjie Chen, Cheng Li, Yanghua Xiao
Long paper, in: **ACL 2024**.

[19] *Light Up the Shadows: Enhance Long-Tail Entity Grounding with Concept-Guided Vision-Language Models*
Yikai Zhang♠, Qianyu He, Xintao Wang, **Siyu Yuan**, Jiaqing Liang, Yanghua Xiao
Long paper, in: **ACL 2024** (Findings).

[20] TaskBench: Benchmarking Large Language Models for Task Automation
Yongliang Shen*, Kaitao Song*, Xu Tan, Wenqi Zhang, Kan Ren, **Siyu Yuan**, Weiming Lu, Dongsheng Li, Yueting Zhuang
Long paper, in: **ICLR 2024 Workshop on LLM Agents**.

[21] Translate Meanings, Not Just Words: IdiomKB's Role in Optimizing Idiomatic Translation with Language Models
Shuang Li♠, Jiangjie Chen, **Siyu Yuan**, Xinyi Wu, Hao Yang, Shimin Tao, Yanghua Xiao
Long paper, in: **AAAI 2024**.

[22] Contextual Information and Commonsense Based Prompt for Emotion Recognition in Conversation
Jingjie Yi, Deqing Yang, **Siyu Yuan**, Kaiyan Cao, Zhiyao Zhang and Yanghua Xiao
Long paper, in: **ECML PKDD 2022**.

Academic Service

- Program Committee/Reviewer of EMNLP (2022-), APIN (2022-), ACL (2023-), ICLR (2024), COLM (2024), and some corresponding workshops

Mentoring

[Knowledge Cutoff: Feb 2024]

- Ruihan Yang, Ph.D. student at Fudan University [4] — June 2023 - Present
- Yikai Zhang, Ph.D. student at Fudan University [8] [19] [4] — May 2023 - Present
- Shuang Li, M.S. student at Fudan University [21] — Jan 2023 - Present
- Nianqi Li, M.S. student at Fudan University [4] — Mar 2024 - Present
- Xinfeng Yuan, M.S. student at Fudan University [17] [16] — Sep 2023 - Present
- Zhijun Xu, M.S. student at Fudan University [3] — Sep 2023 - Present
- Tianhe Lin, M.S. student at Fudan University [16] — Sep 2023 - Present
- Yuhan Cui, B.S. student at Fudan University [16] — Sep 2023 - Present
- Linjie Chen, B.S. student at Fudan University [3] — Sep 2023 - Present
- Xuyang Ge, B.S. student at Fudan University → M.S. student at Fudan University [9][10] — Jul 2022 - Dec 2023